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## GRAVITATION

## KEY CONCEPT

## The discovery of the law of gravitation

The way the law of universal gravitation was discovered is often considered the paradigm of modern scientific technique. The major steps involved were

- The hypothesis about planetary motion given by **Nicolaus Copernicus (1473–1543)**.
- The careful experimental measurements of the positions of the planets and the Sun by **Tycho Brahe (1546–1601)**.
- Analysis of the data and the formulation of empirical laws by **Johannes Kepler (1571–1630)**.
- The development of a general theory by **Isaac Newton (1642–1727)**.

## Newton's law of Gravitation

It states that every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and is inversely proportional to the square of the distance between them.

$$F \propto m_1 m_2 \text{ and } F \propto \frac{1}{r^2} \text{ so } F \propto \frac{m_1 m_2}{r^2}$$

$$\therefore F = -\frac{Gm_1 m_2}{r^2} \hat{r} \quad [G = \text{Universal gravitational constant}]$$

**Note :** This formula is only applicable for spherical symmetric masses or point masses.

## Vector form of Newton's law of Gravitation :

Let  $\vec{r}_{12}$  = Displacement vector from  $m_1$  to  $m_2$

$\vec{r}_{21}$  = Displacement vector from  $m_2$  to  $m_1$

$\vec{F}_{21}$  = Gravitational force exerted on  $m_2$  by  $m_1$

$\vec{F}_{12}$  = Gravitational force exerted on  $m_1$  by  $m_2$

$$\vec{F}_{12} = -\frac{Gm_1 m_2}{r_{21}^2} \hat{r}_{21} = -\frac{Gm_1 m_2}{r_{21}^3} \vec{r}_{21}$$

Negative sign shows that :

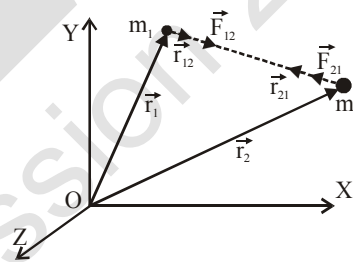
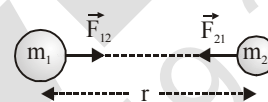
- The direction of  $\vec{F}_{12}$  is opposite to that  $\hat{r}_{21}$
- The gravitational force is attractive in nature

$$\text{Similarly } \vec{F}_{21} = -\frac{Gm_1 m_2}{r_{12}^2} \hat{r}_{12} \text{ or } \vec{F}_{21} = -\frac{Gm_1 m_2}{r_{12}^3} \vec{r}_{12} \Rightarrow \vec{F}_{12} = -\vec{F}_{21}$$

The gravitational force between two bodies are equal in magnitude and opposite in direction.

## Gravitational constant "G"

- Gravitational constant is a scalar quantity.
- **Unit :** **SI :**  $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$   
**CGS :**  $6.67 \times 10^{-8} \text{ dyne-cm}^2/\text{g}^2$   
**Dimensions :**  $\text{M}^{-1}\text{L}^3\text{T}^{-2}$
- Its value is same throughout the universe, G does not depend on the nature and size of the bodies, it also does not depend upon nature of the medium between the bodies.



- Its value was first find out by the scientist "**Henry Cavendish**" with the help of "Torsion Balance" experiment.
- Value of  $G$  is small therefore gravitational force is weaker than electrostatic and nuclear forces.

**Ex.** Two particles of masses 1 kg and 2 kg are placed at a separation of 50 cm. Assuming that the only forces acting on the particles are their mutual gravitation, find the initial acceleration of heavier particle.

**Sol.** Force exerted by one particle on another  $F = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 1 \times 2}{(0.5)^2} = 5.3 \times 10^{-10} \text{ N}$

$$\text{Acceleration of heavier particle} = \frac{F}{m_2} = \frac{5.3 \times 10^{-10}}{2} = 2.65 \times 10^{-10} \text{ ms}^{-2}$$

This example shows that gravitation is very weak but only this force keep bind our solar system and also this universe, all galaxies and other interstellar system.

**Ex.** Two stationary particles of masses  $M_1$  and  $M_2$  are at a distance 'd' apart. A third particle lying on the line joining the particles, experiences no resultant gravitational forces. What is the distance of this particle from  $M_1$ .

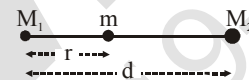
**Sol.** The force on  $m$  towards  $M_1$  is  $F_1 = \frac{GM_1m}{r^2}$

$$\text{The force on } m \text{ towards } M_2 \text{ is } F_2 = \frac{GM_2m}{(d-r)^2}$$

According to question net force on  $m$  is zero i.e.  $F_1 = F_2$

$$\Rightarrow \frac{GM_1m}{r^2} = \frac{GM_2m}{(d-r)^2} \Rightarrow \left( \frac{d-r}{r} \right)^2 = \frac{M_2}{M_1}$$

$$\Rightarrow \frac{d}{r} - 1 = \frac{\sqrt{M_2}}{\sqrt{M_1}} \Rightarrow r = d \left[ \frac{\sqrt{M_1}}{\sqrt{M_1} + \sqrt{M_2}} \right]$$



**Ex.** Three particles, each of mass  $m$ , are situated at the vertices of an equilateral triangle of side 'a'. The only forces acting on the particles are their mutual gravitational forces. It is desired that each particle moves in a circle while maintaining their original separation 'a'. Determine the initial velocity that should be given to each particle and time period of circular motion.

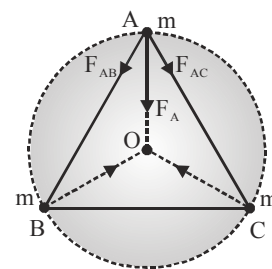
**Sol.** The resultant force on particle at A due to other two particles is

$$F_A = \sqrt{F_{AB}^2 + F_{AC}^2 + 2F_{AB}F_{AC}\cos 60^\circ} = \sqrt{3} \frac{Gm^2}{a^2} \dots (i) \quad \left[ \because F_{AB} = F_{AC} = \frac{Gm^2}{a^2} \right]$$

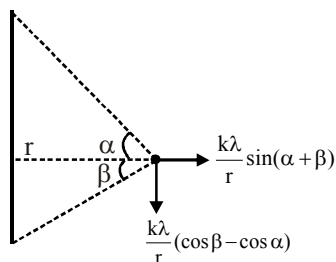
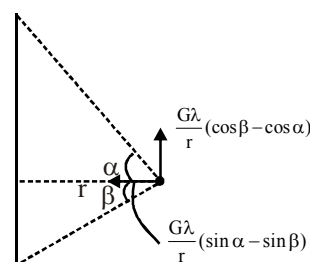
$$\text{Radius of the circle } r = \frac{2}{3} \times a \sin 60^\circ = \frac{a}{\sqrt{3}}$$

If each particle is given a tangential velocity  $v$ , so that  $F$  acts as the centripetal force,

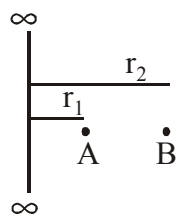
$$\text{Now } \frac{mv^2}{r} = \sqrt{3} \frac{mv^2}{a} \dots (ii)$$





**(3) Uniform linear charge****Uniform linear mass**

( $\lambda$  = mass per unit length)

**(4) Infinite Linear charge**

$$(a) \quad E = \frac{2K\lambda}{r}$$

$$(b) \quad V_B - V_A = -2K\lambda \ln \frac{r_2}{r_1}$$

**(5) Infinite Sheet of charge**

$$E = \frac{\sigma}{2\epsilon_0}$$

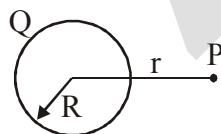
\* Notice gravitational force is always attractive and hence gravitational potential is always -ve. (for a repulsive force potential is positive). This can be explained from the sign of  $W_{\text{ext}}$  in moving the test charge from  $\infty$  to the point under consideration.

\*\* Since  $\vec{g}$  points from B towards A potential increases as we move from A to B. Just like electric potential gravitational potential also increases opposite to field direction.

**(6) Uniformly charged hollow sphere**

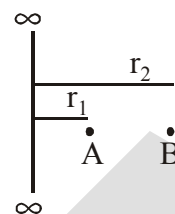
Charge  $Q$ , radius  $R$   
distance of field point from center  $r$

**Case I  $r > R$**



$$E = \frac{kQ}{r^2}$$

$$V = \frac{kQ}{r}$$

**Infinite linear mass**

$$g = \frac{2G\lambda}{r}$$

$$V_B - V_A = 2G\lambda \ln \left( \frac{r_2}{r_1} \right)$$

**Infinite Sheet of mass**

$$g = \frac{\sigma}{2} \times 4\pi G = 2\pi G\sigma$$

( $\sigma$  = mass per unit area)

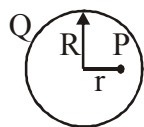
**Hollow sphere of uniform mass**

Mass  $M$ , radius  $R$   
distance of field point from center  $r$

$$g = \frac{GM}{r^2}$$

$$V_G = -\frac{GM}{r}$$

Case II  $r < R$



$$E = 0$$

$$V = \frac{kQ}{r}$$

- (7) **Electrostatics self energy of uniformly charged thin spherical shell.**

$$U = \frac{KQ^2}{2R}$$

- (8) **Uniformly charged solid sphere**  
mass  $M$ , radius  $R$

$$E = \frac{kQ}{r^2}, r > R$$

$$\frac{kQr}{R^3}, r < R$$

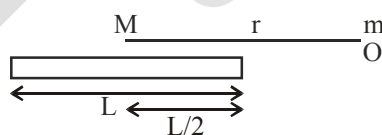
$$V = \frac{kQ}{r}, r > R$$

$$\frac{kQ}{2R^3}(3R^3 - r^2), r > R$$

- (9) **Electrostatics self energy of uniformly charged solid sphere.**

$$U = \frac{3}{5} \frac{KQ^2}{R}$$

**Ex.** Find gravitational force between the point mass & the rod of uniform mass.



**Ans.** 
$$\frac{GMm}{\left(r + \frac{1}{2}\right)\left(r - \frac{1}{2}\right)}$$

**Sol.** 
$$dm = \frac{M}{L} dx$$

$$dF = \frac{G(dm)m}{x^2}$$

$$g = 0$$

$$V_G = -\frac{GM}{R}$$

**Gravitational self energy of uniform thin spherical shell.**

$$U = \frac{GM^2}{2R}$$

**Uniformly solid sphere**  
mass  $M$ , radius  $R$

$$g = \frac{GM}{r^2}, r > R$$

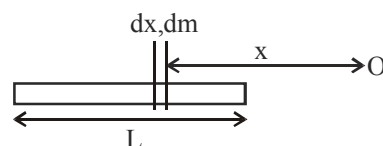
$$\frac{GM}{R^3}r, r < R$$

$$V_a = -\frac{GM}{r}, r > R$$

$$-\frac{GM}{2R^3}(3R^3 - r^2), r > R$$

**Gravitational self energy of uniform solid sphere.**

$$U = \frac{3}{5} \frac{GM^2}{R}$$



$$\int_0^F dF = \frac{GMm}{L} \int_{x_1=\left(r-\frac{L}{2}\right)}^{x_2=\left(r+\frac{L}{2}\right)} \frac{dx}{x^2}$$

$$F = \frac{GMm}{L} \left[ \frac{1}{x} \right]_{x_2}^{x_1} = \frac{GMm}{L} \frac{L}{\left(r-\frac{L}{2}\right)\left(r+\frac{L}{2}\right)} = \frac{GMm}{\left(r-\frac{L}{2}\right)\left(r+\frac{L}{2}\right)}$$

**Ex.** A thin rod of mass  $M$  and length  $L$  is bent in a semicircle as shown in figure.

(a) What is its gravitational force (both magnitude and direction) on a particle with mass  $m$  at  $O$ , the centre of curvature?

(b) What would be the force on  $m$  if the rod is, in the form of a complete circle?

**Sol.** (a) Considering an element of rod of length  $d\ell$  as shown in figure and treating it as a point of mass  $(M/L)d\ell$  situated at a distance  $R$  from  $P$ , the gravitational force due to this element on the particle will be

$$dF = \frac{Gm(M/L)(Rd\theta)}{R^2} \text{ along OP [as } d\ell = Rd\theta \text{]}$$

So the component of this force along  $x$  and  $y$ -axis will be

$$dF_x = dF \cos\theta = \frac{GmM}{LR} \cos\theta d\theta; dF_y = dF \sin\theta = \frac{GmM}{LR} \sin\theta d\theta$$

$$\text{So that } F_x = \frac{GmM}{LR} \int_0^\pi \cos\theta d\theta = \frac{GmM}{LR} [\sin\theta]_0^\pi = 0$$

$$\text{and } F_y = \frac{GmM}{LR} \int_0^\pi \sin\theta d\theta = \frac{GmM}{LR} [-\cos\theta]_0^\pi = \frac{2\pi GmM}{L^2} \left[ \text{as } R = \frac{L}{\pi} \right]$$

$$\text{So } F = \sqrt{F_x^2 + F_y^2} = F_y = \frac{2\pi GmM}{L^2} \text{ [as } F_x \text{ is zero]}$$

i.e., the resultant force is along the  $y$ -axis and has magnitude  $(2\pi GmM/L^2)$

(b) If the rod was bent into a complete circle,

$$F_x = \frac{GmM}{LR} \int_0^{2\pi} \cos\theta d\theta = 0 \text{ and also } F_y = \frac{GmM}{LR} \int_0^{2\pi} \sin\theta d\theta = 0$$

i.e., the resultant force on  $m$  at  $O$  due to the ring is zero.

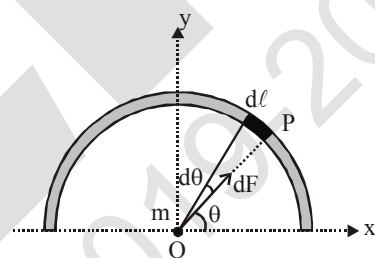
**Ex.** Find ratio of gravitational field on the surface of two planets which are of uniform mass density & have radius  $R_1$  &  $R_2$  if

(a) They are of same mass

(b) They are of same density

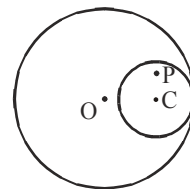
$$\text{Ans. (a) } \frac{g_1}{g_2} = \frac{R_2^2}{R_1^2} \quad (b) \frac{g_1}{g_2} = \frac{R_1}{R_2}$$

$$\text{Sol. } g = \frac{GM}{R^2} = \frac{G \frac{4}{3} \pi R^3 \rho}{R^2} = \left( \frac{4\pi G R}{3} \right) \rho$$



**Ex.** A uniform solid sphere of density  $\rho$  and radius  $R$  has a spherical cavity of radius  $r$  inside it as shown. Find gravitation field at

- (a) O  
(b) C  
(c) P (prove that field inside cavity is uniform)



**Ans.** (a)  $\frac{k\left(\frac{4}{3}\pi r^3\rho\right)}{(OC)^2}\hat{CO}$  (b)  $\frac{4\pi G\rho\overline{OC}}{3}$  (c)  $\frac{4}{3}\pi G\rho\overline{O_1O_2}$

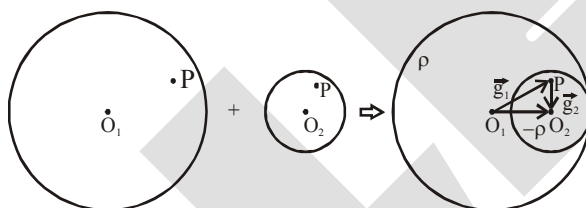
**Sol.**  $\vec{g}$  = gravitational field at any point inside sphere

$$\vec{g} = \frac{GM}{R^3} \vec{r}$$

$$= \frac{G}{R^3} \frac{4}{3} \pi R^3 \rho \vec{r}$$

$$\vec{g} = \frac{4}{3} \pi G \rho \vec{r}$$

Let the sphere with cavity is formed by superimposing it with a small sphere of density  $(-\rho)$  as shown



Resultant field  $\vec{g} = \vec{g}_1 + \vec{g}_2$

$$= \left( \frac{4}{3} \pi G \rho \right) \overline{O_1P} + \left( \frac{4}{3} \pi G \rho \right) \overline{PO_2}$$

$$= \frac{4}{3} \pi G \rho \overline{O_1O_2} \quad \left[ \overline{O_1O_2} = \overline{OC} \right]$$

It is independent of position of point inside cavity

At O  $\vec{g} = \vec{g}_1 + \vec{g}_2$

$$= 0 + \frac{GM}{(\overline{CO})^2} (\widehat{CO})$$

$$= \frac{G \frac{4}{3} \pi r^3 \rho}{(\overline{CO})^2} \widehat{CO}$$

$$= \frac{\left( \frac{4}{3} \pi G r^3 \rho \right)}{(\overline{CO})^2} \widehat{CO}$$

**Acceleration Due to Gravity (g)**

Gravitational Force  $F_g = ma$  if  $R_e$  = Radius of Earth,  $M_e$  = Mass of Earth.

$$\text{then } \frac{GM_e m}{R_e^2} = ma_g \Rightarrow a_g = g = \frac{GM_e}{R_e^2} \quad (GM_e = gR_e^2) \dots (i)$$

• In form of density  $g = \frac{GM_e}{R_e^2} = \frac{G}{R_e^2} \times \frac{4}{3} \pi R_e^3 \times \rho$

$$\therefore g = \frac{4}{3} \pi G R_e \rho \quad \dots (ii)$$

If  $\rho$  is constant then  $g \propto R_e$

**Variation in Acceleration due gravity****(a) Due to Altitude (height)**

$$\frac{g_h}{g} = \frac{R_e^2}{(R_e + h)^2}$$

By binomial expansion  $\left(1 + \frac{h}{R_e}\right)^{-2} \approx \left(1 - \frac{2h}{R_e}\right)$

[If  $h \ll R_e$ , then higher power terms are negligible]  $\therefore g_h = g \left[1 - \frac{2h}{R_e}\right]$

**Ex.** Two equal masses  $m$  and  $m$  are hung from a balance whose scale pans differ in vertical height by 'h'. Determine the error in weighing in terms of density of the Earth  $\rho$ .

**Sol.**  $g_h = g \left[1 - \frac{2h}{R_e}\right]$ ,  $W_2 - W_1 = mg_2 - mg_1 = 2mg \left[\frac{h_1}{R_e} - \frac{h_2}{R_e}\right] = 2m \frac{GM}{R_e^2} \times \frac{h}{R_e} \left[\because g = \frac{GM}{R_e^2} \text{ \& } h_1 - h_2 = h\right]$

Error in weighing =  $W_2 - W_1 = 2mG \frac{4}{3} \pi R_e^3 \rho \frac{h}{R_e^3} = \frac{8\pi}{3} G m \rho h$

**(b) Due to depth :**

Assuming density of Earth remains same throughout. At depth  $d$  inside the Earth :

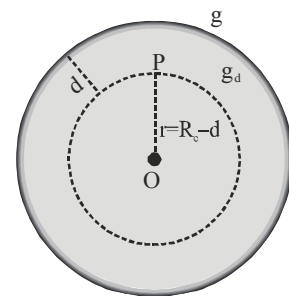
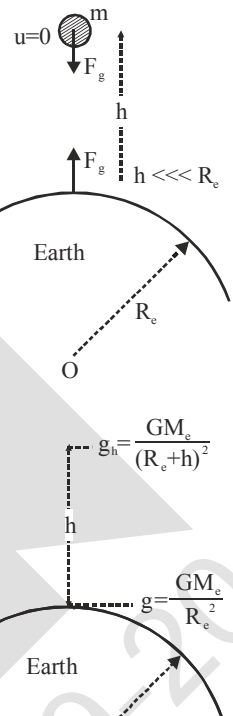
$$g_d = g \left[1 - \frac{d}{R_e}\right] \text{ valid for any depth}$$

Decrement in  $g$  with depth =  $\Delta g_d = g - g_d = g - g \left[1 - \frac{d}{R_e}\right]$

$$\therefore \frac{\Delta g_d}{g} = \frac{d}{R_e}$$

**Ex.** At which depth from Earth surface, acceleration due to gravity is decreased by 1%

**Sol.**  $\frac{\Delta g_d}{g} = \frac{d}{R_e} \Rightarrow \frac{1}{100} = \frac{d}{6400} \therefore d = 64 \text{ km}$

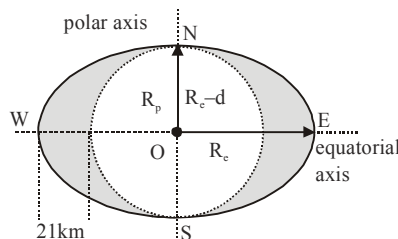


**(c) Due to shape of the Earth**

$$R_p < R_e$$

$$\therefore g_e < g_p$$

$$\Rightarrow \text{by putting the values } g_p - g_e = 0.02 \text{ m/s}^2$$



**(d) Due to Rotation of the Earth**

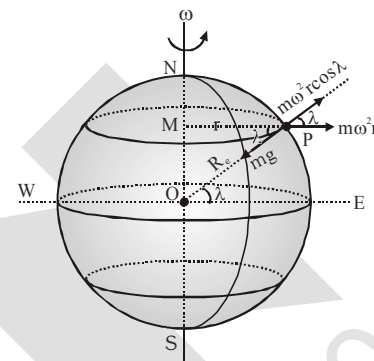
$$g' = g - R_e \omega^2 \cos^2 \lambda$$

If latitude angle  $\lambda = 0$ . It means at equator.  $g'_{\min} = g_e = g - R_e \omega^2$

If latitude angle  $\lambda = 90^\circ$ . It means at poles.  $g'_{\max} = g_p = g \Rightarrow g_p > g_e$

Change in "g" only due to rotation  $\Delta g_{\text{rot}} = g_p - g_e = 0.03 \text{ m/s}^2$

$$\Delta g_{\text{total}} = g_p - g_e = (0.05 \text{ m/s}^2) \rightarrow \begin{cases} 0.02 \text{ m/s}^2 & \text{(due to shape)} \\ 0.03 \text{ m/s}^2 & \text{(due to rotation)} \end{cases}$$



**Weightlessness**

State of the free fall  $\left( \vec{a} = -\frac{GM}{r^2} \vec{r} \right)$  is called state of weightlessness. If a body is in a satellite (which does not produce its own gravity) orbiting the Earth at a height  $h$  above its surface then

$$\text{True weight} = mg_h = \frac{mGM}{(R+h)^2} = \frac{mg}{\left(1 + \frac{h}{R}\right)^2}$$

$$\text{Apparent weight} = m(g_h - a)$$

$$\text{but } a = \frac{v_0^2}{r} = \frac{GM}{r^2} = \frac{GM}{(R+h)^2} = g_h \Rightarrow \text{Apparent weight} = m(g_h - g_h) = 0$$

**Note :** The condition of weightlessness can be overcome by creating artificial gravity by rotating the satellite in addition to its revolution.

**Escape speed ( $v_e$ )**

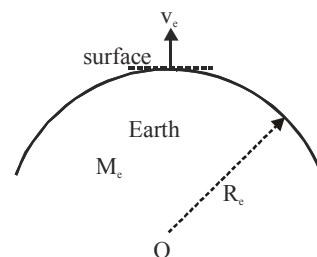
Minimum speed required for an object at Earth's surface so that it just escapes the Earth's gravitational field.

**Escape energy**

Minimum energy given to a particle in form of kinetic energy so that it can just escape from Earth's gravitational field.

$$\text{Escape energy} = \frac{GM_e m}{R_e} \text{ (-ve of PE of Earth's surface)}$$

$$\text{Escape energy} = \text{Kinetic Energy} \Rightarrow \frac{GM_e m}{R_e} = \frac{1}{2} m v_e^2 \Rightarrow v_e = \sqrt{\frac{2GM_e}{R_e}}$$





(b) The initial minimum speed of the body to reach smaller star is  $K\sqrt{\frac{GM}{a}}$ . Find the value of K:

**Sol.** From conservation of mechanical energy.

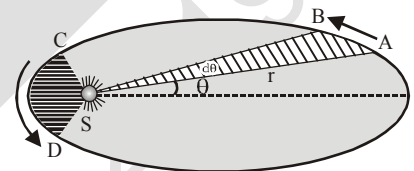
$$\begin{aligned}\frac{1}{2}mv_{\min}^2 &= \text{Potential energy of body at P} - \text{Potential energy of body at larger star} \\ &= \left[ -\frac{GMm}{r_1} - \frac{16GMm}{r_2} \right] - \left[ -\frac{GMm}{(10a-2a)} - \frac{16GMm}{2a} \right] \\ &= \frac{3\sqrt{5}}{2} \left[ \sqrt{\frac{GM}{a}} \right]\end{aligned}$$

### Kepler's Laws

Kepler found important regularities in the motion of the planets. These regularities are known as 'Kepler's three laws of planetary motion'.

- (a) **First Law (Law of Orbits)** : All planets move around the Sun in elliptical orbits, having the Sun at one focus of the orbit.
- (b) **Second Law (Law of Areas)** : A line joining any planet to the Sun sweeps out equal areas in equal times, that is, the areal speed of the planet remains constant.

According to the second law, when the planet is nearest the Sun, then its speed is maximum and when it is farthest from the Sun, then its speed is minimum. In figure if a planet moves from A to B in a given time-interval, and from C to D in the same time-interval, then the areas ASB and CSD will be equal.



$$\frac{dA}{dt} = \frac{J}{2m} \quad \dots(iii)$$

Now, the areal speed  $dA/dt$  of the planet is constant, according to Kepler's second law. Therefore, according to eq. (iii), the angular momentum  $J$  of the planet is also constant, that is, the angular momentum of the planet is conserved. Thus, Kepler's second law is equivalent to conservation of angular momentum.

- (c) **Third Law : (Law of Periods)** : The square of the period of revolution (time of one complete revolution) of any planet around the Sun is directly proportional to the cube of the semi-major axis of its elliptical orbit.

$$T^2 \propto a^3$$

So it is clear through this rule that the farthest planet from the Sun has largest period of revolution. The period of revolution of the closest planet Mercury is 88 days, while that of the farthest dwarf planet Pluto is 248 years.

### Satellite motion

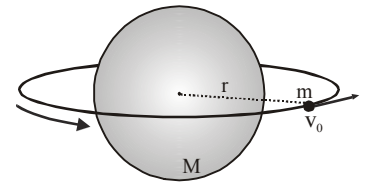
A light body revolving round a heavier body due to gravitational attraction, is called satellite. Earth is a satellite of the Sun while Moon is satellite of Earth.

**Orbital velocity ( $v_0$ )** : A satellite of mass  $m$  moving in an orbit of radius  $r$  with speed  $v_0$  then required centripetal force is provided by gravitation.

$$F_{cp} = F_g \Rightarrow \frac{mv_0^2}{r} = \frac{GMm}{r^2} \Rightarrow v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{(R_e + h)}} \quad (r = R_e + h)$$

For a satellite very close to the Earth surface  $h \ll R_e \therefore r = R_e$

$$v_0 = \sqrt{\frac{GM}{R_e}} = \sqrt{gR_e} = 8 \text{ km/s}$$



- If a body is taken at some height from Earth and given horizontal velocity of magnitude 8 km/sec then the body becomes satellite of Earth.
- $v_0$  depends upon : Mass of planet, Radius of circular orbit of satellite,  $g$  (at planet), Density of planet
- If orbital velocity of a near by satellite becomes  $\sqrt{2} v_0$  (or increased by 41.4%, or K.E. is doubled) then the satellite escapes from gravitational field of Earth.

**Time Period of a Satellite**  $T = \frac{2\pi r}{v_0} = \frac{2\pi r^{3/2}}{\sqrt{GM}} = \frac{2\pi r^{3/2}}{R\sqrt{g}} \Rightarrow T^2 = \frac{4\pi^2}{GM} r^3 \Rightarrow T^2 \propto r^3 (r = R + h)$

**For Geostationary Satellite**  $T = 24 \text{ hr}$ ,  $h = 36,000 \text{ km} \approx 6 R_e$  ( $r \approx 7 R_e$ ),  $v_0 = 3.1 \text{ km/s}$

**For Near by satellite**  $v_0 = \sqrt{\frac{GM_e}{R_e}} \approx 8 \text{ km/s}$

$$T_{Ns} = 2\pi \sqrt{\frac{R_e}{g}} = 84 \text{ minute} = 1 \text{ hour } 24 \text{ minute} = 1.4 \text{ hr} = 5063 \text{ s}$$

In terms of density  $T_{Ns} = \frac{2\pi(R_e)^{1/2}}{(G \times 4/3 \pi R_e \times \rho)^{1/2}} = \sqrt{\frac{3\pi}{G\rho}}$

Time period of near by satellite only depends upon density of planet.

**For Moon**  $h_m = 380,000 \text{ km}$  and  $T_m = 27 \text{ days}$

$$v_{om} = \frac{2\pi(R_e + h)}{T_m} = \frac{2\pi(386400 \times 10^3)}{27 \times 24 \times 60 \times 60} \approx 1.04 \text{ km/sec.}$$

**Energies of a Satellite** Kinetic energy  $K.E. = \frac{1}{2} mv_0^2 = \frac{GMm}{2r} = \frac{L^2}{2mr^2}$

Potential energy  $P.E. = -\frac{GMm}{r} = -mv_0^2 = -\frac{L^2}{mr^2}$

Total mechanical energy T.E. = P.E. + K.E. =  $-\frac{mv_0^2}{2} = -\frac{GMm}{2r} = -\frac{L^2}{2mr^2}$

### Essential Condition's for Satellite Motion

- Centre of satellite's orbit coincide with centre of Earth.
- Plane of orbit of satellite is passing through centre of Earth.

### Special Points about Geo-Stationary Satellite

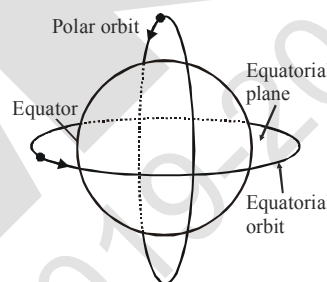
- All three essential conditions for satellite motion should be followed.
- It rotates in equatorial plane.
- Its height from Earth surface is 36000 km. ( $\sim 6R_e$ )
- Its angular velocity and time period should be same as that of Earth.
- Its rotating direction should be same as that of Earth (West to East).
- Its orbit is called parking orbit and its orbital velocity is 3.1 km./sec.
- Maximum latitude at which message can be received by geostationary satellite is

$$\theta = \cos^{-1} \left( \frac{R_e}{R_e + h} \right)$$

- The area of earth's surface covered by geostationary satellite is  $S = \Omega R_e^2 = \frac{2\pi h R_e^2}{R_e + h}$

### • Polar Satellite (Sun – synchronous satellite)

It is that satellite which revolves in polar orbit around Earth. A polar orbit is that orbit whose angle of inclination with equatorial plane of Earth is  $90^\circ$  and a satellite in polar orbit will pass over both the north and south geographic poles once per orbit. Polar satellites are Sun-synchronous satellites. Every location on Earth lies within the observation of polar satellite twice each day. The polar satellites are used for getting the cloud images, atmospheric data, ozone layer in the atmosphere and to detect the ozone hole over Antarctica.



Only the equatorial orbits are stable for a satellite. For any satellite to orbit around in a stable orbit, it must move in such an orbit so that the centre of Earth lies at the centre of the orbit.

### Binding energy

Total mechanical energy (potential + kinetic) of a closed system is negative. The modulus of this total mechanical energy is known as the binding energy of the system. This is the energy due to which system is closed or different parts of the system are bound to each other.

#### Binding energy of satellite (system)

$$\text{B.E.} = -\text{T.E.} \quad \text{B.E.} = \frac{1}{2}mv_0^2 = \frac{GMm}{2r} = \frac{L^2}{2mr^2} \quad \text{Hence B.E.} = \text{K.E.} = -\text{T.E.} = \frac{-\text{P.E.}}{2}$$

### Work done in Changing the Orbit of Satellite

$$W = \text{Change in mechanical energy of system but } E = \frac{-GMm}{2r} \text{ so } W = E_2 - E_1 = \frac{GMm}{2} \left( \frac{1}{r_1} - \frac{1}{r_2} \right)$$

**Ex.** A satellite moves eastwards very near the surface of the Earth in equatorial plane with speed ( $v_0$ ). Another satellite moves at the same height with the same speed in the equatorial plane but westwards. If  $R$  = radius of the Earth and  $\omega$  be its angular speed of the Earth about its own axis. Then find the approximate difference in the two time period as observed on the Earth.

$$\text{Sol. } T_{\text{west}} = \frac{2\pi R}{v_0 + R\omega} \text{ and } T_{\text{east}} = \frac{2\pi R}{v_0 - R\omega} \Rightarrow \Delta T = T_{\text{east}} - T_{\text{west}} = 2\pi R \left[ \frac{2R\omega}{v_0^2 - R^2\omega^2} \right] = \frac{4\pi\omega R^2}{v_0^2 - R^2\omega^2}$$

**Ex.** An artificial satellite (mass  $m$ ) of a planet (mass  $M$ ) revolves in a circular orbit whose radius is  $n$  times the radius  $R$  of the planet. In the process of motion, the satellite experiences a slight resistance due to cosmic dust. Assuming the force of resistance on satellite to depend on velocity as  $F = av^2$  where ' $a$ ' is a constant, calculate how long the satellite will stay in the space before it falls onto the planet's surface.

**Sol.** Air resistance  $F = -av^2$ , where orbital velocity  $v = \sqrt{\frac{GM}{r}}$

$r$  = the distance of the satellite from planet's centre  $\Rightarrow F = -\frac{GMa}{r}$

The work done by the resistance force  $dW = Fdx = Fvdt = \frac{GMa}{r} \sqrt{\frac{GM}{r}} dt = \frac{(GM)^{3/2}}{r^{3/2}} a dt$  ....(i)

The loss of energy of the satellite =  $dE \therefore \frac{dE}{dr} = \frac{d}{dr} \left[ -\frac{GMm}{2r} \right] = \frac{GMm}{2r^2} \Rightarrow dE = \frac{GMm}{2r^2} dr$  ... (ii)

Since  $dE = -dW$  (work energy theorem)  $-\frac{GMm}{2r^2} dr = \frac{(GM)^{3/2}}{r^{3/2}} a dt$

$$\Rightarrow t = -\frac{m}{2a\sqrt{GM}} \int_{nR}^R \frac{dr}{\sqrt{r}} = \frac{m\sqrt{R}(\sqrt{n}-1)}{a\sqrt{GM}} = (\sqrt{n}-1) \frac{m}{a\sqrt{gR}}$$

**Ex.** Two satellites  $S_1$  and  $S_2$  revolve round a planet in coplanar circular orbits in the same sense. Their periods of revolution are 1 h and 8h respectively. The radius of the orbit of  $S_1$  is  $10^4$  km. When  $S_2$  is closest to  $S_1$ , find (a) the speed of  $S_2$  relative to  $S_1$  and (b) the angular speed of  $S_2$  as observed by an astronaut in  $S_1$ .

**Sol.** Let the mass of the planet be  $M$ , that of  $S_1$  be  $m_1$  and  $S_2$  be  $m_2$ . Let the radius of the orbit of  $S_1$  be  $R_1 (= 10^4 \text{ km})$  and of  $S_2$  be  $R_2$ .

Let  $v_1$  and  $v_2$  be the linear speeds of  $S_1$  and  $S_2$  with respect to the planet. Figure shows the situation. As the square of the time period is proportional to the cube of the radius.

$$\left( \frac{R_2}{R_1} \right)^3 = \left( \frac{T_2}{T_1} \right)^2 = \left( \frac{8h}{1h} \right)^2 = 64$$

$$\text{or, } \frac{R_2}{R_1} = 4$$

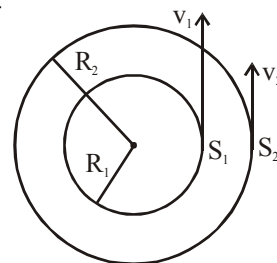
$$\text{or, } R_2 = 4R_1 = 4 \times 10^4 \text{ m.}$$

Now the time period of  $S_1$  is 1 h. So,

$$\frac{2\pi R_1}{v_1} = 1h$$

$$\text{or, } v_1 = \frac{2\pi R_1}{1h} = 2\pi \times 10^4 \text{ km/h}$$

$$\text{similarly, } v_2 = \frac{2\pi R_2}{8h} = \pi \times 10^4 \text{ km/h}$$



- (a) At the closest separation, they are moving in the same direction. Hence the speed of  $S_2$  with respect to  $S_1$  is  $|v_2 - v_1| = \pi \times 10^4$  km/h.
- (b) As seen from  $S_1$ , the satellite  $S_2$  is at a distance  $R_2 - R_1 = 3 \times 10^4$  km at the closest separation. Also it is moving at  $\pi \times 10^4$  km/h in a direction perpendicular to the line joining them. Thus, the angular speed of  $S_2$  as observed by  $S_1$  is

$$\omega = \frac{\pi \times 10^4 \text{ km/h}}{3 \times 10^4 \text{ km/h}} = \frac{\pi}{3} \text{ rad/h.}$$

### Conditions for different trajectory

For a body being projected tangentially from above earth's surface, say at a distance  $r$  from earth's center, the trajectory would depend on the velocity of projection  $v$ .

#### Velocity

1. velocity,  $v < \sqrt{\frac{GM}{r} \left( \frac{2R}{r+R} \right)}$

2.  $\sqrt{\frac{GM}{r}} > v > \sqrt{\frac{GM}{r} \left( \frac{2R}{r+R} \right)}$

3. Velocity is equal to the critical velocity

of the orbit,  $v = \sqrt{\frac{GM_e}{r}}$

4. Velocity is between the critical and escape velocity of the orbit

$$\sqrt{\frac{2GM_e}{r}} > v > \sqrt{\frac{GM_e}{r}}$$

5.  $v = v_{\text{esc}} = \sqrt{\frac{2GM_e}{r}}$

6.  $v > v_{\text{esc}} = \sqrt{\frac{2GM_e}{r}}$

#### Orbit

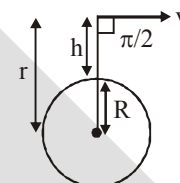
Body returns to earth

Body acquires an elliptical orbit with earth as the far-focus w.r.t. the point of projection.  
Circular orbit with radius  $r$

Body acquires an elliptical orbit with earth as the near focus w.r.t. the point of projection.

Body just escapes earth's gravity, along a parabolic path.

Body escape earth's gravity along a hyperbolic path.

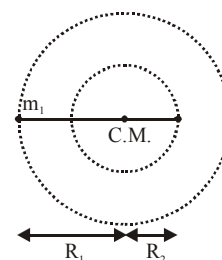


### Binary star system

Figure shows two particles moving due to mutually attractive gravitational force about center of mass. Since there is no external force CM of system remains fixed and time period of revolution must be same.

Both bodies have comparable mass and both are moving in circular orbit centre of mass as shown in diagram

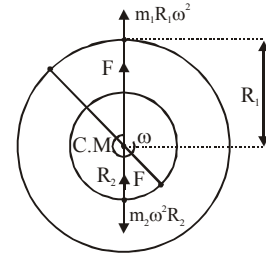
$$\omega = \sqrt{\frac{G(m_1 + m_2)}{R^3}}$$



Angular momentum of the system about centre of mass.

$$L = \left( \frac{m_1 m_2}{m_1 + m_2} \right) R^2 \omega$$

$$\text{Kinetic energy} = \frac{1}{2} \left( \frac{m_1 m_2}{m_1 + m_2} \right) R^2 \omega^2$$



### The time period of a simple pendulum of infinite length.

In deriving the formula  $T_0 = 2\pi \sqrt{\left(\frac{L}{g}\right)}$  we have assumed that length of the pendulum  $L$  is much less than the radius of the earth  $R$  so that 'g' always remains vertical. However, if length of pendulum is comparable of the radius of earth, 'g' will not remain vertical but will be directed towards the centre of the earth.

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{1}{g \left[ \frac{1}{L} + \frac{1}{R} \right]}} \quad (< T_0)$$

From this expression it is clear that :

- (a) If  $L \ll R$ ,  $(1/L) \gg (1/R)$  so  $T = 2\pi \sqrt{\left(\frac{L}{g}\right)}$  which is expected.
- (b) If  $L \gg R \rightarrow \infty$   $(1/L) \ll (1/R)$  so

$$T = 2\pi \sqrt{\frac{R}{g}} = 2\pi \sqrt{\frac{6.4 \times 10^6}{10}}$$

$$= 800 \times 2\pi \text{ sec} \approx 83.8 \text{ minute}$$

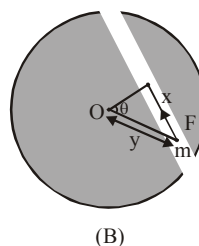
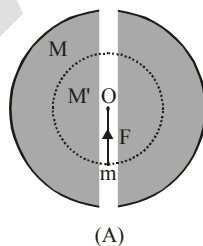
And it is also the maximum time period which an oscillating simple pendulum can have.

- (c) If  $L$  is comparable to  $R$  (say  $L = R$ ),

$$T = 2\pi \sqrt{\frac{R}{2g}} \approx 1 \text{ hour}.$$

### Motion of a ball in a tunnel through the earth :

**Case I :** If the tunnel is along a diameter and the ball is released from the surface. The ball executes SHM.



$$\text{so that } T = 2\pi \sqrt{\frac{R^3}{GM}} ; T = 2\pi \sqrt{\left(\frac{R}{g}\right)}$$

Which is same as that of a simple pendulum of infinite length and is equal to 84.6 minute.

**Case II :** If the tunnel is along a chord and ball is released from the surface. The motion is SHM with the same time period.

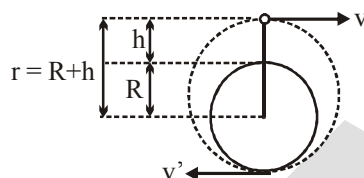
**Gravitational pressure :**

A uniform sphere has a mass  $M$  and radius  $R$ . The pressure  $p$  inside the sphere, caused by gravitational compression, as a function of the distance  $r$  from its centre can be found to be  $p = \frac{3}{8} (1 - r^2/R^2) \gamma M^2/\pi R^4$ .

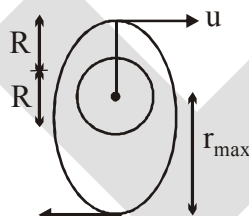
**Launching of an artificial satellite around earth**

**Ex.** A satellite is launched tangentially from a height  $h$  above earth's surface as shown.

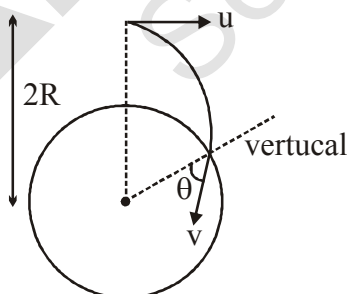
I. Find minimum launch speed ' $v$ ' so that it just touches the earth's surface



II. If  $h = R$  and satellite is launched tangentially with speed  $= \sqrt{\frac{3GM}{5R}}$  find the maximum distance of satellite from earth's center



(III) If  $h = R$  and satellite is launched tangentially with a speed  $u = \sqrt{\frac{GM}{7R}}$ . Find the angle w.r.t. vertical at which the satellite will crash on earth's surface.



**Sol.**

(I) Angular momentum conservation

$$mv(R + h) = mv'R$$

Energy conservation

$$\frac{-GMm}{R+h} + \frac{1}{2}mv^2 = -\frac{GMm}{R} + \frac{1}{2}mv'^2$$

$$\text{Solving, } v = \sqrt{\frac{2GMR}{r(R+r)}}$$

(II) Angular momentum conservation  $mu \cdot 2R = mvr_{\max}$

Energy conservation :

$$\frac{-GMm}{2R} + \frac{1}{2}mu^2 = \frac{-GMm}{r_{\max}} + \frac{1}{2}mv^2$$

$$\Rightarrow GMm \left( \frac{1}{2R} - \frac{1}{r_{\max}} \right) = \frac{1}{2}mu^2 \left( 1 - \left( \frac{2R}{r_{\max}} \right)^2 \right)$$

$$\Rightarrow GMm \left( \frac{1}{2R} - \frac{1}{r_{\max}} \right) = \frac{3GM}{10R} \left( 1 - \frac{4R^2}{r_{\max}^2} \right)$$

$$\Rightarrow 2r_{\max}^2 - 10Rr_{\max} + 12R^2 = 0$$

$$\Rightarrow (r_{\max} - 2R)(r_{\max} - 3R) = 0$$

$$\Rightarrow r_{\max} = 3R$$

(III) Energy conservation

$$\frac{1}{2}mu^2 - \frac{GMm}{2R} = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

$$\Rightarrow \frac{1}{2}m(v^2 - u^2) = \frac{GMm}{2R}$$

$$v^2 = u^2 + \frac{GM}{R} \Rightarrow v = \sqrt{\frac{8GM}{7R}}$$

Angular momentum conservation

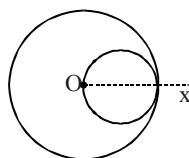
$$mu \cdot 2R = mvR \sin \theta$$

$$\Rightarrow \sin \theta = \frac{2u}{v}$$

$$\Rightarrow \theta = 45^\circ$$

## EXERCISE (S-1)

1. A particle is fired vertically from the surface of the earth with a velocity  $k v_e$ , where  $v_e$  is the escape velocity and  $k < 1$ . Neglecting air resistance and assuming earth's radius as  $R_e$ . Calculate the height to which it will rise from the surface of the earth.
2. Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.
3. An object is projected vertically upward from the surface of the earth of mass  $M$  with a velocity such that the maximum height reached is eight times the radius  $R$  of the earth. Calculate:
  - (i) the initial speed of projection
  - (ii) the speed at half the maximum height.
4. A satellite close to the earth is in orbit above the equator with a period of rotation of 1.5 hours. If it is above a point  $P$  on the equator at some time, it will be above  $P$  again after time \_\_\_\_\_.
5. A satellite is moving in a circular orbit around the earth. The total energy of the satellite is  $E = -2 \times 10^5 \text{ J}$ . The amount of energy to be imparted to the satellite to transfer it to a circular orbit where its potential energy is  $U = -2 \times 10^5 \text{ J}$  is equal to \_\_\_\_\_.
6. A satellite of mass  $m$  is orbiting the earth in a circular orbit of radius  $r$ . It starts losing energy due to small air resistance at the rate of  $C \text{ J/s}$ . Then the time taken for the satellite to reach the earth is \_\_\_\_\_.
7. A pair of stars rotates about a common center of mass. One of the stars has a mass  $M$  which is twice as large as the mass  $m$  of the other. Their centres are at a distance  $d$  apart,  $d$  being large compared to the size of either star.
  - (a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of  $d, m, G$ .
  - (b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio  $L_m / L_M$ .
  - (c) Compare the kinetic energies of the two stars by calculating the ratio  $K_m / K_M$ .
8. A sphere of radius  $R$  has its centre at the origin. It has a uniform mass density  $\rho_0$  except that there is a spherical hole of radius  $r = R/2$  whose centre is at  $x = R/2$  as in fig. (a) Find gravitational field at points on the axis for  $|x| > R$  (b) Show that the gravitational field inside the hole is uniform, find its magnitude and direction.



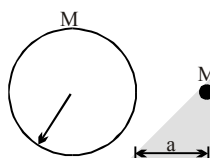


### EXERCISE (O-1)

1. If the distance between the centres of Earth and Moon is  $D$  and mass of Earth is 81 times that of Moon. At what distance from the centre of Earth gravitational field will be zero?

(A)  $\frac{D}{2}$  (B)  $\frac{2D}{3}$  (C)  $\frac{4D}{5}$  (D)  $\frac{9D}{10}$

2. A particle of mass  $M$  is at a distance  $a$  from surface of a thin spherical shell of equal mass and having radius  $a$ .



- (A) Gravitational field and potential both are zero at centre of the shell.  
 (B) Gravitational field is zero not only inside the shell but at a point outside the shell also.  
 (C) Inside the shell, gravitational field alone is zero.  
 (D) Neither gravitational field nor gravitational potential is zero inside the shell.
3. A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre
- (A) increases  
 (B) decreases  
 (C) remains same  
 (D) during the compression increases then returns at the previous value.
4. Let  $\omega$  be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles in absence of rotation of earth. An object weighed at the equator gives the same reading as a reading taken at a depth  $d$  below earth's surface at a pole ( $d \ll R$ ) The value of  $d$  is

(A)  $\frac{\omega^2 R^2}{g}$  (B)  $\frac{\omega^2 R^2}{2g}$  (C)  $\frac{2\omega^2 R^2}{g}$  (D)  $\frac{\sqrt{Rg}}{g}$

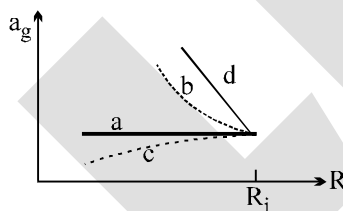
5. If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of  $g$  the same?

(A)  $1/25$  (B)  $1/5$  (C)  $1/\sqrt{5}$  (D)  $5$

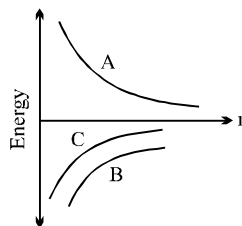
6. The mass and diameter of a planet are twice those of earth. What will be the period of oscillation of a pendulum on this planet if it is a seconds pendulum on earth?

(A)  $\sqrt{2}$  second (B)  $2\sqrt{2}$  seconds (C)  $\frac{1}{\sqrt{2}}$  second (D)  $\frac{1}{2\sqrt{2}}$  second

7. Two identical satellites are at the heights  $R$  and  $7R$  from the Earth's surface. Then which of the following statement is incorrect. ( $R$  = radius of the Earth)
- (A) Ratio of total energy of both is 5  
 (B) Ratio of kinetic energy of both is 4  
 (C) Ratio of potential energy of both 4  
 (D) Ratio of total energy of both is 4 and ratio of magnitude of potential to kinetic energy is 2
8. A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is  $V$ . Due to the rotation of planet about its axis the acceleration due to gravity  $g$  at equator is  $1/2$  of  $g$  at poles. The escape velocity of a particle on the pole of planet in terms of  $V$  is
- (A)  $V_e = 2V$                       (B)  $V_e = V$                       (C)  $V_e = V/2$                       (D)  $V_e = \sqrt{3} V$
9. The escape velocity for a planet is  $v_e$ . A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be
- (A)  $v_e$                       (B)  $\frac{v_e}{\sqrt{2}}$                       (C)  $\frac{v_e}{2}$                       (D) 0
10. A (nonrotating) star collapses onto itself from an initial radius  $R_i$  with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration  $a_g$  on the surface of the star as a function of the radius of the star during the collapse?



- (A) a                      (B) b                      (C) c                      (D) d
11. A satellite of mass  $m$ , initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is
- (A)  $\frac{\sqrt{3}}{4} mgR$                       (B)  $\frac{1}{2} mgR$                       (C)  $\frac{1}{4} mgR$                       (D)  $\frac{3}{4} mgR$
12. The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



- (A) A shows the kinetic energy, B the total energy and C the potential energy of the system.  
 (B) C shows the total energy, B the kinetic energy and A the potential energy of the system.  
 (C) C and A are kinetic and potential energies respectively and B is the total energy of the system.  
 (D) A and B are kinetic and potential energies and C is the total energy of the system.

13. A satellite of mass  $5M$  orbits the earth in a circular orbit. At one point in its orbit, the satellite explodes into two pieces, one of mass  $M$  and the other of mass  $4M$ . After the explosion the mass  $M$  ends up travelling in the same circular orbit, but in opposite direction. After explosion the mass  $4M$  is :-  
 (A) In a circular orbit  
 (B) unbound  
 (C) elliptical orbit  
 (D) data is insufficient to determine the nature of the orbit.
14. A satellite can be in a geostationary orbit around earth at a distance  $r$  from the centre. If the angular velocity of earth about its axis doubles, a satellite can now be in a geostationary orbit around earth if its distance from the centre is :-  
 (A)  $\frac{r}{2}$  (B)  $\frac{r}{2\sqrt{2}}$  (C)  $\frac{r}{(4)^{1/3}}$  (D)  $\frac{r}{(2)^{1/3}}$
15. An earth satellite is moved from one stable circular orbit to another larger and stable circular orbit. The following quantities increase for the satellite as a result of this change:-  
 (A) gravitational potential energy (B) angular velocity  
 (C) linear orbital velocity (D) centripetal acceleration
16. Satellites A and B are orbiting around the earth in orbits of ratio  $R$  and  $4R$  respectively. The ratio of their areal velocities is :  
 (A)  $1 : 2$  (B)  $1 : 4$  (C)  $1 : 8$  (D)  $1 : 16$
17. The fractional change in the value of free-fall acceleration  $g$  for a particle when it is lifted from the surface to an elevation  $h$  ( $h \ll R$ ) is  
 (A)  $\frac{h}{R}$  (B)  $\frac{2h}{R}$  (C)  $-\frac{2h}{R}$  (D)  $-\frac{h}{R}$
18. If suddenly the gravitational force of attraction between earth and a satellite revolving around it becomes zero, then the satellite will- [AIEEE-2002]  
 (A) continue to move in its orbit with same velocity  
 (B) move tangentially to the original orbit with same velocity  
 (C) become stationary in its orbit  
 (D) move towards the earth
19. The time period of a satellite of earth is 5 hours. If the separation between the centre of earth and the satellite is increased to 4 times the previous value, the new time period will become- [AIEEE-2003]  
 (A) 10 h (B) 80 h (C) 40 h (D) 20 h
20. A communications Earth satellite  
 (A) goes round the earth from east to west  
 (B) can be in the equatorial plane only  
 (C) can be vertically above any place on the earth  
 (D) goes round the earth from west to east

21. If a satellite orbits as close to the earth's surface as possible,  
(A) its speed is maximum  
(B) time period of its rotation is minimum  
(C) the total energy of the 'earth plus satellite' system is minimum  
(D) the total energy of the 'earth plus satellite' system is maximum
22. For a satellite to orbit around the earth, which of the following must be true?  
(A) It must be above the equator at some time  
(B) It cannot pass over the poles at any time  
(C) Its height above the surface cannot exceed 36,000 km  
(D) Its period of rotation must be  $> 2\pi\sqrt{R/g}$  where R is radius of earth
23. In elliptical orbit of a planet, as the planet moves from apogee position to perigee position,

**Column-I**

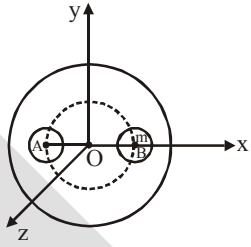
- (A) Speed of planet  
(B) Distance of planet from centre of Sun  
(C) Potential energy  
(D) Angular momentum about centre of Sun

**Column-II**

- (P) Remains same  
(Q) Decreases  
(R) Increases  
(S) Can not say

### EXERCISE (O-2)

- A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit, with their centres at A  $(-2, 0, 0)$  and B  $(2, 0, 0)$  respectively, are taken out of the solid leaving behind spherical cavities as shown in figure. Then



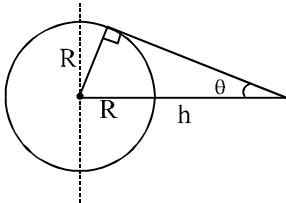
(A) The gravitational field due to this object at the origin is zero  
 (B) The gravitational field at the point B  $(2, 0, 0)$  is zero  
 (C) The gravitational potential is the same at all points of circle  $y^2 + z^2 = 36$   
 (D) The gravitational potential is the same at all points on the circle  $y^2 + z^2 = 4$
- A planet of mass  $m$  is in an elliptical orbit about the sun ( $m \ll M_{\text{sun}}$ ) with an orbital period  $T$ . If  $A$  be the area of orbit, then its angular momentum would be :

(A)  $\frac{2mA}{T}$                       (B)  $mAT$                       (C)  $\frac{mA}{2T}$                       (D)  $2mAT$
- A planet revolves about the sun in elliptical orbit. The areal velocity  $\left(\frac{dA}{dt}\right)$  of the planet is  $4.0 \times 10^{16} \text{ m}^2/\text{s}$ . The least distance between planet and the sun is  $2 \times 10^{12} \text{ m}$ . Then the maximum speed of the planet in  $\text{km/s}$  is :

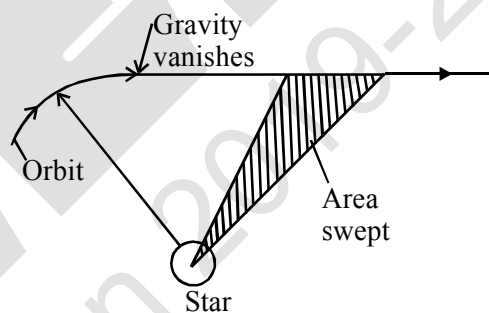
(A) 10                      (B) 20                      (C) 40                      (D) None of these
- The Sun travels in approximately circular orbit of radius  $R$  around the center of the galaxy and completes one revolution in time  $T$ . The Earth also revolves around the Sun in time  $t$ . Assume orbit of the Earth to be a circle of radius  $r$  ( $r \ll R$ ) and whole mass of the galaxy centered on its center. By using only these given informations, find an expression for the ratio of the mass of the galaxy to that of the Sun.

(A)  $\left(\frac{R}{r}\right)^3 \left(\frac{t}{T}\right)^2$                       (B)  $\left(\frac{R}{r}\right)^3 \left(\frac{T}{t}\right)^2$                       (C)  $\left(\frac{R}{r}\right)^2 \left(\frac{t}{T}\right)^3$                       (D)  $\left(\frac{R}{r}\right)^2 \left(\frac{T}{t}\right)^3$
- A geostationary satellite is at a height  $h$  above the surface of earth. If earth radius is  $R$

(A) The minimum colatitude on earth upto which the satellite can be used for communication is  $\sin^{-1}(R/R+h)$ .  
 (B) The maximum latitudes on earth upto which the satellite can be used for communication is  $\cos^{-1}(R/R+h)$ .  
 (C) The area on earth escaped from this satellite is given as  $2\pi R^2 (1 + \sin\theta)$   
 (D) The area on earth escaped from this satellite is given as  $2\pi R^2 (1 + \cos\theta)$

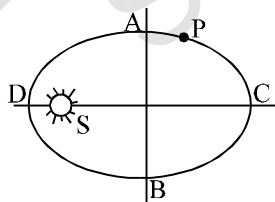


6. When a satellite in a circular orbit around the earth enters the atmospheric region, it encounters small air resistance to its motion. Then  
 (A) its kinetic energy increases  
 (B) its kinetic energy decreases  
 (C) its angular momentum about the earth decreases  
 (D) its period of revolution around the earth increases
7. Two satellites  $s_1$  &  $s_2$  of equal masses revolve in the same sense around a heavy planet in coplanar circular orbit of radii  $R$  &  $4R$   
 (A) the ratio of period of revolution of  $s_1$  &  $s_2$  is  $1 : 8$ .  
 (B) their velocities are in the ratio  $2 : 1$   
 (C) their angular momentum about the planet are in the ratio  $2 : 1$   
 (D) the ratio of angular velocities of  $s_2$  w.r.t.  $s_1$  when all three are in the same line is  $9 : 5$ .
8. A double star is a system of two stars of masses  $m$  and  $2m$ , rotating about their centre of mass only under their mutual gravitational attraction. If  $r$  is the separation between these two stars then their time period of rotation about their centre of mass will be proportional to :  
 (A)  $r^{3/2}$  (B)  $r$  (C)  $m^{1/2}$  (D)  $m^{-1/2}$
9. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the **CORRECT** statement(s) :  
 (A) Newton's first law is obeyed on planet after gravity vanishes  
 (B) Kepler's law of areas is obeyed only till the planet is in gravity of star  
 (C) Kepler's law of areas is obeyed even after gravity vanishes  
 (D) Angular momentum of planet about centre of star is conserved through out its motion



### Paragraph for Question No. 10 and 11

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



10. If  $t_1$  is the time taken by the planet to travel along ACB and  $t_2$  the time along BDA, then  
 (A)  $t_1 = t_2$  (B)  $t_1 > t_2$   
 (C)  $t_1 < t_2$  (D) nothing can be concluded
11. If  $U$  is the potential energy and  $K$  kinetic energy then  $|U| > |K|$  at  
 (A) Only D (B) Only C (C) both D & C (D) neither D nor C

### EXERCISE (JM)

1. The height at which the acceleration due to gravity becomes  $\frac{g}{9}$  (where  $g$  = the acceleration due to gravity on the surface of the earth) in terms of  $R$ , the radius of the earth, is :- [AIEEE - 2009]

(1)  $\frac{R}{2}$  (2)  $\sqrt{2}R$  (3)  $2R$  (4)  $\frac{R}{\sqrt{2}}$

2. Two bodies of masses  $m$  and  $4m$  are placed at a distance  $r$ . The gravitational potential at a point on the line joining them where the gravitational field is zero is :- [AIEEE - 2011]

(1)  $-\frac{6Gm}{r}$  (2)  $-\frac{9Gm}{r}$  (3) zero (4)  $-\frac{4Gm}{r}$

3. Two particles of equal mass ' $m$ ' go around a circle of radius  $R$  under the action of their mutual gravitational attraction. The speed of each particle with respect to their centre of mass is:- [AIEEE-2011]

(1)  $\sqrt{\frac{Gm}{R}}$  (2)  $\sqrt{\frac{Gm}{4R}}$  (3)  $\sqrt{\frac{Gm}{3R}}$  (4)  $\sqrt{\frac{Gm}{2R}}$

4. The mass of a spaceship is 1000 kg. It is to be launched from the earth's surface out into free space. The value of ' $g$ ' and ' $R$ ' (radius of earth) are  $10 \text{ m/s}^2$  and 6400 km respectively. The required energy for this work will be :- [AIEEE-2012]

(1)  $6.4 \times 10^{10}$  Joules (2)  $6.4 \times 10^{11}$  Joules (3)  $6.4 \times 10^8$  Joules (4)  $6.4 \times 10^9$  Joules

5. What is the minimum energy required to launch a satellite of mass  $m$  from the surface of a planet of mass  $M$  and radius  $R$  in a circular orbit at an altitude of  $2R$  ? [JEE-Main 2013]

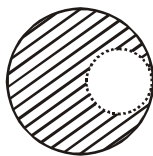
(1)  $\frac{5GmM}{6R}$  (2)  $\frac{2GmM}{3R}$  (3)  $\frac{GmM}{2R}$  (4)  $\frac{GmM}{3R}$

6. Four particles, each of mass  $M$  and equidistant from each other, move along a circle of radius  $R$  under the action of their mutual gravitational attraction. The speed of each particle is :

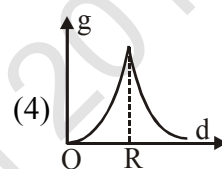
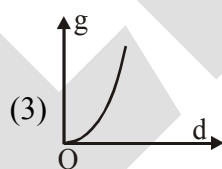
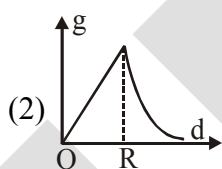
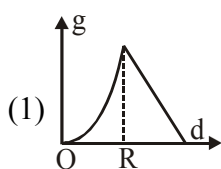
[JEE-Main 2014]

(1)  $\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$  (2)  $\frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$  (3)  $\sqrt{\frac{GM}{R}}$  (4)  $\sqrt{2\sqrt{2}\frac{GM}{R}}$

7. From a solid sphere of mass  $M$  and radius  $R$ , a spherical portion of radius  $\frac{R}{2}$  is removed, as shown in the figure. Taking gravitational potential  $V = 0$  at  $r = \infty$ , the potential at the centre of the cavity thus formed is : ( $G = \text{gravitational constant}$ ) [JEE-Main 2015]

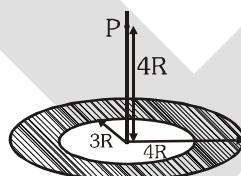


- (1)  $\frac{-2GM}{3R}$       (2)  $\frac{-2GM}{R}$       (3)  $\frac{-GM}{2R}$       (4)  $\frac{-GM}{R}$
8. A satellite is revolving in a circular orbit at a height ' $h$ ' from the earth's surface (radius of earth  $R$  ;  $h \ll R$ ). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to : (Neglect the effect of atmosphere). [JEE-Main 2016]
- (1)  $\sqrt{gR}(\sqrt{2} - 1)$       (2)  $\sqrt{2gR}$       (3)  $\sqrt{gR}$       (4)  $\sqrt{gR/2}$
9. The variation of acceleration due to gravity  $g$  with distance  $d$  from centre of the earth is best represented by ( $R = \text{Earth's radius}$ ) :- [JEE-Main 2017]



## EXERCISE (JA)

- Gravitational acceleration on the surface of a planet is  $\frac{\sqrt{6}}{11}g$ , where  $g$  is the gravitational acceleration on the surface of the earth. The average mass density of the planet is  $\frac{2}{3}$  times that of the Earth. If the escape speed on the surface of the earth is taken to be  $11 \text{ kms}^{-1}$ , the escape speed on the surface of the planet in  $\text{kms}^{-1}$  will be [IIT-JEE 2010]
- A binary star consists of two stars A (mass  $2.2 M_s$ ) and B (mass  $11 M_s$ ), where  $M_s$  is the mass of the sun. They are separated by distance  $d$  and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is :- [IIT-JEE 2010]
- A thin uniform annular disc (see figure) of mass  $M$  has outer radius  $4R$  and inner radius  $3R$ . The work required to take a unit mass from point P on its axis to infinity is [IIT-JEE 2010]



- (A)  $\frac{2GM}{7R}(4\sqrt{2}-5)$  (B)  $-\frac{2GM}{7R}(4\sqrt{2}-5)$
- (C)  $\frac{GM}{4R}$  (D)  $\frac{2GM}{5R}(\sqrt{2}-1)$
- A satellite is moving with a constant speed  $V$  in a circular orbit about the earth. An object of mass ' $m$ ' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of its ejection, the kinetic energy of the object is :- [IIT-JEE 2011]
- (A)  $\frac{1}{2}mV^2$  (B)  $mV^2$  (C)  $\frac{3}{2}mV^2$  (D)  $2mV^2$
- Two spherical planets P and Q have the same uniform density  $\rho$ , masses  $M_p$  and  $M_Q$ , and surface areas  $A$  and  $4A$ , respectively. A spherical planet R also has uniform density  $\rho$  and its mass is  $(M_p + M_Q)$ . The escape velocities from the planets P, Q and R, are  $V_p$ ,  $V_Q$  and  $V_R$ , respectively. Then [IIT-JEE 2012]
- (A)  $V_Q > V_R > V_p$  (B)  $V_R > V_Q > V_p$  (C)  $V_R/V_p = 3$  (D)  $V_p/V_Q = 1/2$

6. A planet of radius  $R = \frac{1}{10} \times$  (radius of Earth) has the same mass density as Earth. Scientists dig a well of depth  $\frac{R}{5}$  on it and lower a wire of the same length and of linear mass density  $10^{-3} \text{ kg m}^{-1}$  into it. If the wire is not touching anywhere, the force applied at the top of the wire by a person holding it in place is (take the radius of Earth  $= 6 \times 10^6 \text{ m}$  and the acceleration due to gravity on Earth is  $10 \text{ ms}^{-2}$ )

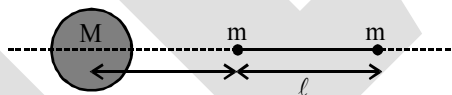
[JEE-Advance 2014]

- (A) 96 N (B) 108 N (C) 120 N (D) 150 N

7. A bullet is fired vertically upwards with velocity  $v$  from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is  $1/4^{\text{th}}$  of its value at the surface of the planet. If the escape velocity from the planet is  $v_{\text{esc}} = v\sqrt{N}$ , then the value of  $N$  is (ignore energy loss due to atmosphere)

[JEE-Advance 2015]

8. A large spherical mass  $M$  is fixed at one position and two identical point masses  $m$  are kept on a line passing through the centre of  $M$  (see figure). The point masses are connected by a rigid massless rod of length  $\ell$  and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to  $M$  is at a distance  $r = 3\ell$  from  $M$ , the tension in the rod is zero for  $m = k\left(\frac{M}{288}\right)$ . The value of  $k$  is: [JEE-Advance 2015]



9. A spherical body of radius  $R$  consists of a fluid of constant density and is in equilibrium under its own gravity. If  $P(r)$  is the pressure at  $r$  ( $r < R$ ), then the correct option(s) is(are) :- [JEE-Advance 2015]

- (A)  $P(r = 0) = 0$  (B)  $\frac{P(r = 3R/4)}{P(r = 2R/3)} = \frac{63}{80}$   
 (C)  $\frac{P(r = 3R/5)}{P(r = 2R/5)} = \frac{16}{21}$  (D)  $\frac{P(r = R/2)}{P(r = R/3)} = \frac{20}{27}$

10. A rocket is launched normal to the surface of the Earth, away from the Sun, along the line joining the sun and the Earth. The Sun is  $3 \times 10^5$  times heavier than the Earth and is at a distance  $2.5 \times 10^4$  times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is  $v_e = 11.2 \text{ km s}^{-1}$ . The minimum initial velocity ( $v_s$ ) required for the rocket to be able to leave the Sun-Earth system is closest to (Ignore the rotation and revolution of the Earth and the presence of any other planet)

[JEE-Advance 2017]

- (A)  $v_s = 22 \text{ km s}^{-1}$  (B)  $v_s = 72 \text{ km s}^{-1}$   
 (C)  $v_s = 42 \text{ km s}^{-1}$  (D)  $v_s = 62 \text{ km s}^{-1}$

11. A planet of mass  $M$ , has two natural satellites with masses  $m_1$  and  $m_2$ . The radii of their circular orbits are  $R_1$  and  $R_2$  respectively. Ignore the gravitational force between the satellites. Define  $v_1$ ,  $L_1$ ,  $K_1$  and  $T_1$  to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and  $v_2$ ,  $L_2$ ,  $K_2$  and  $T_2$  to be the corresponding quantities of satellite 2. Given  $m_1/m_2 = 2$  and  $R_1/R_2 = 1/4$ , match the ratios in List-I to the numbers in List-II.

[JEE-Advance 2018]

**List-I**

P.  $\frac{v_1}{v_2}$

Q.  $\frac{L_1}{L_2}$

R.  $\frac{K_1}{K_2}$

S.  $\frac{T_1}{T_2}$

**List-II**

1.  $\frac{1}{8}$

2. 1

3. 2

4. 8

(A)  $P \rightarrow 4$  ;  $Q \rightarrow 2$  ;  $R \rightarrow 1$  ;  $S \rightarrow 3$

(C)  $P \rightarrow 2$  ;  $Q \rightarrow 3$  ;  $R \rightarrow 1$  ;  $S \rightarrow 4$

(B)  $P \rightarrow 3$  ;  $Q \rightarrow 2$  ;  $R \rightarrow 4$  ;  $S \rightarrow 1$

(D)  $P \rightarrow 2$  ;  $Q \rightarrow 3$  ;  $R \rightarrow 4$  ;  $S \rightarrow 1$

## ANSWER KEY

## EXERCISE (S-1)

1. Ans.  $\frac{R_e k^2}{1-k^2}$       2. Ans.  $h = \frac{\sqrt{5}-1}{2} R$       3. Ans. (i)  $\frac{4}{3} \sqrt{\frac{Gm}{R}}$ , (ii)  $\frac{2}{3} \sqrt{\frac{2Gm}{5R}}$

4. Ans. 1.6 hours if it is rotating from west to east, 24/17 hours if it is rotating from east to west

5. Ans.  $1 \times 10^5 \text{ J}$       6. Ans.  $t = \frac{GMm}{2C} \left( \frac{1}{R_e} - \frac{1}{r} \right)$

7. Ans. (a)  $T = \frac{2\pi d^{3/2}}{\sqrt{3Gm}}$ , (b) 2, (c) 2

8. Ans.  $\vec{g} = +\frac{\pi G \rho_0 R^3}{6} \left[ \frac{1}{(x-(R/2))^2} - \frac{8}{x^2} \right] \hat{i}$ ,  $\vec{g} = -\frac{2\pi G \rho_0 R}{3} \hat{i}$

## EXERCISE (S-2)

1. Ans.  $2R, 3R[3-\sqrt{3}]$       2. Ans. (a)  $h = \left( \frac{\sqrt{7}}{2} + 1 \right) R$ , (b)  $1.125 R$

3. Ans. (a)  $\sqrt{\frac{GM}{r}}$ ; (b)  $r\sqrt{2}$ ; (c)  $\sqrt{\frac{2GM}{r}}$       4. Ans.  $T = 3 \times 10^{-2} \text{ N}$

5. Ans.  $\frac{1}{4} K^2 G \pi (R^4 - r^4)$       6. Ans.  $7/3$       7. Ans. 3

8. Ans. (a)  $\frac{2Gm^2}{3R}$  (b)  $4R$       9. Ans.  $\left( 1 - \frac{\sqrt{x^2 - R^2}}{x} \right) 4\pi R^2$

## EXERCISE (O-1)

1. Ans. (D)    2. Ans. (D)    3. Ans. (B)    4. Ans. (A)    5. Ans. (B)    6. Ans. (B)  
 7. Ans. (A)    8. Ans. (A)    9. Ans. (B)    10. Ans. (B)    11. Ans. (D)    12. Ans. (D)  
 13. Ans. (B)    14. Ans. (C)    15. Ans. (A)    16. Ans. (A)    17. Ans. (C)    18. Ans. (B)  
 19. Ans. (C)    20. Ans. (B,D)    21. Ans. (A,B,C)    22. Ans. (A, D)  
 23. Ans. (A)-R, (B)-Q, (C)-Q (D)-P

**EXERCISE (O-2)**

1. Ans. (A,C,D)    2. Ans. (A)    3. Ans. (C)    4. Ans. (A)    5. Ans. (A,B,C)  
 6. Ans. (A,C)    7. Ans. (A,B,D)    8. Ans. (A,D)    9. Ans. (A,C,D)    10. Ans. (B)  
 11. Ans. (C)

**EXERCISE (JM)**

1. Ans. (3)    2. Ans. (2)    3. Ans. (2)    4. Ans. (1)    5. Ans. (1)    6. Ans. (2)  
 7. Ans. (4)    8. Ans. (1)    9. Ans. (2)

**EXERCISE (JA)**

1. Ans. 3    2. Ans. 6    3. Ans. (A)    4. Ans. (B)    5. Ans. (B,D)    6. Ans. (B)  
 7. Ans. 2    8. Ans. 7    9. Ans. (B,C)    10. Ans. (C)    11. Ans. (B)